CTFS - uvod

Vrem. kont. Fourier red

$$F_k = \frac{1}{T_0}\int_{T_0} f(t)\, e^{-jk\frac{2\pi}{T_0}t}\, dt \quad , \forall k \in \mathbb{Z}$$

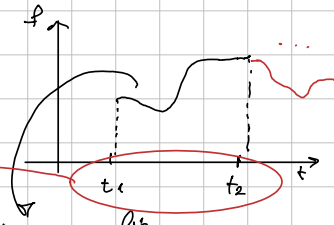
T_0 - osnovni period

koeficijenti CTFS

- postoje kada je definiran i konačan $\forall k \in \mathbb{Z}$ → tada ih nazivamo spektar signala f

REKONSTRUKCIJA signala iz spektra F: → $\{F_k\}_{k\in\mathbb{Z}}$ čine spektar signala (niz brojeva)

$$f(t) = \sum_{k=-\infty}^{\infty} F_k e^{jk\frac{2\pi}{T_0}t} \quad , \forall t \in \mathbb{R}$$



$\forall k$ od prošli put $F_k = \frac{1}{t_2-t_1}\int_{t_1}^{t_2} f(t)\, e^{-jk\frac{2\pi}{t_2-t_1}t}\, dt$

periodičan ako bi se u tom intervalu ponovio taj segment
$\Rightarrow t_2 - t_1 = T_0$

opisan skupom fja

$$f(t) = \left\{ e^{-j\frac{2\pi}{t_2-t_1}kt} \right\}_{k\in\mathbb{Z}}$$

to vrijedi kada bismo „periodizirali" signal

f period. kont. signal; $f: \mathbb{R} \to \mathbb{R}\ (\mathbb{C})$

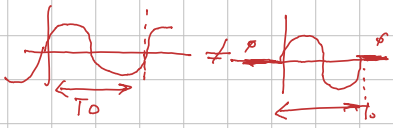
$$f(t+T_0) = f(t) \quad \forall t \in \mathbb{R}$$

→ potpuno isto imamo li periodičan sig. ili signal definiran na samo 1 periodi

$$f(t+T_0) = \sum F_k e^{jk\frac{2\pi}{T_0}(t+T_0)}$$

$$= \sum F_k \cdot e^{jk\frac{2\pi}{T_0}} \cdot \underbrace{e^{jk2\pi}}_{1} = \underline{\underline{\sum F_k \cdot e^{jk\frac{2\pi}{T_0}}}}$$

ALI!



Ako imamo spektar (niz brojeva) → možemo nazad dobiti f

Rekonstrukcija signala: $f(t) = \sum_{k=-\infty}^{\infty} F_k e^{jk\frac{2\pi}{T_0}t} \quad , \forall t \in \mathbb{R}$

⇒ kada F_k postoje i rekonstrukcija konvergira! ⇒ f i F predstavljaju par vremenski kont. F. reda

▶ f i F kada označavamo s: $F = CTFS(f)$

$f = ICTFS(F)$

$|F|$ = amplitudni spektar

$\angle F$ = fazni spektar

CTFS- konvergencija

Signal konačne en.

tada vremenski kont. Fourierov red konvergira u srednjem (energija pogreške na periodi je nula), tj. vrijedi ?

ako vrijedi $\int_{T_0} |f(t)|^2 dt < \infty \Rightarrow \int_0^{T_0} \left| f(t) - \sum_{k=-K}^{K} F_k e^{jk\omega_0 t} \right|^2 dt \to 0$ za $k \to \infty$

Signali koji zadovoljavaju Dirichletove uvjete:

- ako je f periodičan i kontinuiran signal koji je:

a) apsolutno integrabilan preko periode

$$\int_{T_0} |f(t)| dt < \infty$$

b) ima konačan broj maksimuma i minimuma u periodi

c) ima konačan broj diskontinuiteta u periodi

tada vremenski kontinuirani F. red konvergira po točkama, osim na mjestima diskontinuiteta, $[f(t_d^+) \neq f(t_d^-)]$ gdje rekonstruirani signal konvergira prema $\frac{(f(t_d^+) + f(t_d^-))}{2}$

Primjer CTFS:

$$F_k = \frac{1}{T_0} \int_{T_0} f(t) e^{-jk\frac{2\pi}{T_0}t} dt$$

$$= \frac{1}{T_0} \int_0^{\tau} A e^{-jk\frac{2\pi}{T_0}kt} dt = \frac{A}{T_0} \int_0^{\tau} e^{-jk\frac{2\pi}{T_0}t} dt = \frac{-A}{T_0} \cdot \frac{1}{jk\frac{2\pi}{T_0}} \cdot e^{-jk\frac{2\pi}{T_0}t} \Big|_0^{\tau}$$

u τ; $f(\tau) = A$

osim za $k=0$

$$= \frac{A}{\cancel{T_0}} \cdot \frac{j \cdot \cancel{T_0}}{k\, 2\pi} \cdot \left(e^{-jk\frac{2\pi}{T_0} \cdot \tau} - 1 \right) = \frac{A \cdot j}{k 2\pi} \cdot e^{-j\frac{\pi}{T_0}k\tau} \left(e^{-j\frac{\pi}{T_0}k\tau} - e^{+j\frac{\pi}{T_0}k\tau} \right)$$

$$= \frac{-A}{j2k\pi} \cdot e^{-j\frac{\pi}{T_0}k\tau} \left(e^{-j\frac{\pi}{T_0}k\tau} - e^{+j\frac{\pi}{T_0}k\tau} \right) = \frac{A}{k\pi} e^{-j\frac{\pi}{T_0}k\tau} \underbrace{\left(\frac{e^{+j\frac{\pi}{T_0}k\tau} - e^{-j\frac{\pi}{T_0}k\tau}}{2j} \right)}_{\sin}$$

$$* \quad \frac{e^{j\varphi} - e^{-j\varphi}}{2j} = \sin(\varphi)$$

$$= \frac{A}{k\pi} \cdot e^{-j\frac{\pi}{T_0}k\tau} \cdot \sin\left(\frac{\pi}{T_0} k\tau \right)$$

$$\boxed{F_k = \frac{A\tau}{T_0} e^{-j\frac{\pi}{T_0}k\tau} \cdot \frac{\sin\left(\frac{\pi}{T_0}k\tau\right)}{\frac{k\pi\tau}{T_0}}}$$

$*$ za $k=0$

$$F_0 = \frac{A}{T_0} \int_0^{\tau} \underbrace{e^{-j \cdot 0 \cdot \frac{2\pi}{T_0}t}}_{1} dt \Rightarrow \boxed{F_0 = \frac{A\tau}{T_0}}$$

→ linijski spektar: $|F_k|$ $\quad$ * $Z = Z_1 \cdot Z_2 \cdot Z_3$

→ koef F. reda poprimaju kompleksne vrijednosti i skup $\{F_k\}\Big|_{k=-\infty}^{\infty}$

• kompl. oba grafa ⇒ $F_k = |F_k| \cdot e^{j\angle F_k}$ = linijski spektar

$$|F_k| = \begin{cases} \left|\frac{A\tau}{T_0}\right|, & k=0 \\ \left|\frac{A\tau}{T_0}\right| \left|\frac{\sin\left(\frac{\pi}{T_0}k\tau\right)}{\frac{k\pi\tau}{T_0}}\right|, & k = \pm 1, \pm 2 \dots \end{cases}$$

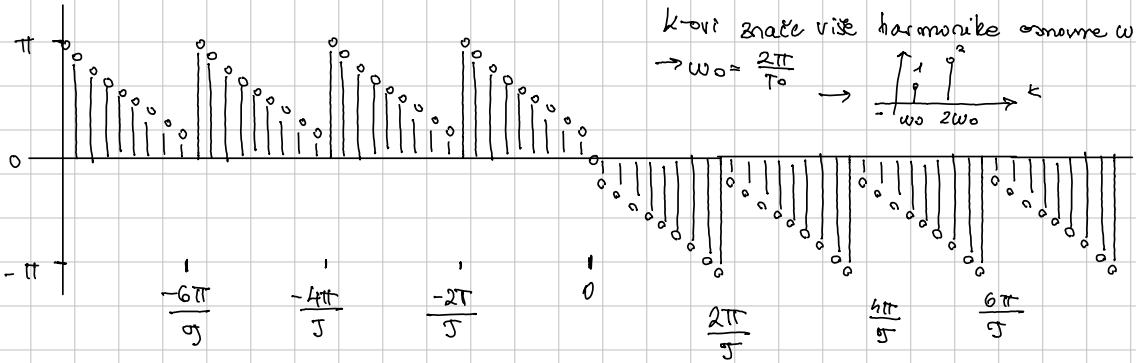
$|Z| = |Z_1|\,|Z_2| \cdot |Z_3|$

$$\angle F_k = \angle A + \underbrace{\angle \frac{\sin\left(\frac{\pi}{T_0}k\tau\right)}{\frac{k\pi\tau}{T_0}}}_{\text{ovo je razlomak pa se od brojnika oduzima nazivnik}} - \frac{k\pi\tau}{T_0}$$

$\measuredangle Z = \measuredangle Z_1 + \measuredangle Z_2 + \measuredangle Z_3$

k-ovi znače više harmonike osnovne ω

→ $\omega_0 = \frac{2\pi}{T_0}$ →

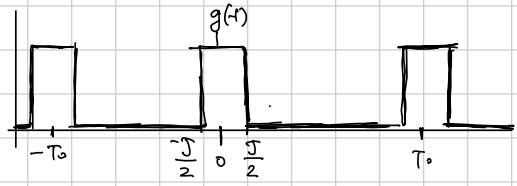


ako gledamo os k → razmak je 1

ako gledamo os frekvencija → razmak je ω_0

- ako je signal paran i realan

⇒ spektar će biti realan



$$G_0 = \frac{A\tau}{T_0}, \quad G_k = \frac{A\tau}{T_0} \frac{\sin\frac{k\pi\tau}{T_0}}{\frac{k\pi\tau}{T_0}}$$ ✓ za razliku od prošlog primjera = realno

⇒ kašnjenje signala za $\tau/2$ rezultira u linearnom faznom pomaku spektra za $-\frac{k\omega_0 T}{2}$, dok amp spektar ostaje nepromijenjen

CTFS - Osnovna svojstva

Parsevalova relacija

- Za periodične signale konačne energije na periodi

*nema limesa jer je periodično

$$P_f = \frac{1}{T_0}\int_{T_0} |f(t)|^2\,dt = \sum_{k=-\infty}^{\infty} |F_k|^2$$

IZVOD:

$$\Rightarrow P_f = \frac{1}{T_0}\int_{T_0} f(t)\,f^*(t)\,dt = \frac{1}{T_0}\int_{T_0} f(t)\sum_k F_k^*\, e^{-j\frac{2\pi}{T_0}kt}\,dt = \sum_k F_k^* \underbrace{\frac{1}{T_0}\int_{T_0} f(t)\,e^{-j\frac{2\pi}{T_0}kt}\,dt}_{F_k}$$

$$\Rightarrow P_f = \sum_k |F_k|^2$$

CTFS realnih signala → za realne parne period. signale f za koje postoje CTFS, spektar signala je realan.

- spektar je konjugirano sim. → $F_{-k} = F^*_k$
- amplitudni spektar je paran, a fazni neparan

$$|F_k| = |F_{-k}| \qquad \angle F_k = -\angle F_{-k}$$

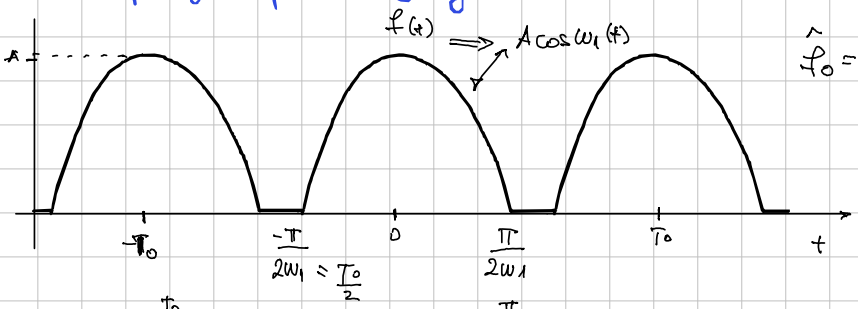
- kompaktni trig oblik: $f(t) = F_0 + \sum_{k=1}^{\infty} 2|F_k|\cos(k\omega_0 t + \angle F_k)$

⇒ najbolja aproksimacija u smislu najmanje energije pogreške

$$\hat{f}_K(t) = \sum_{k=-K}^{K} F_k \cdot e^{jk\frac{2\pi}{T_0}t}$$

je najbolja aprox signala f u smislu min en. greške na prostoru signala razapetim ortonormiranim signalima $\left\{e^{jk\frac{2\pi}{T_0}t}\right\}_{k=-K}^{K}$

CTFS - primjer neprekinutog signala



$$\hat{f}_0 = F_0 \cdot 1 = \frac{A \cdot 2}{T_0 \cdot \omega_1}$$

$$\cos\omega_1 t = \frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2}$$

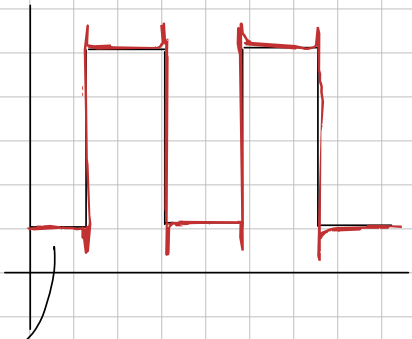
$$F_k = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f(t)\,e^{-jk\omega_0 t}\,dt = \frac{1}{T_0}\int_{-\frac{\pi}{2\omega_1}}^{\frac{\pi}{2\omega_1}} A\cos(\omega_1 t)\,e^{-jk\omega_0 t}\,dt = \frac{2A\omega_1}{T_0(\omega_1^2 - k\omega_0^2)}\cos\left(\frac{\pi\omega_0}{2\omega_1}k\right)$$

za $k = 0, \pm 1, \pm 2 \ldots$

Gibbsova pojava

$K = 1$

$K = 35$



da je $K = 390$, super bi se poklapao na horizont. ali bila bi veća nadvišenja

$\Rightarrow$ rekonstrukcijom nećemo dobiti identičan polazni (2 ε slajd)

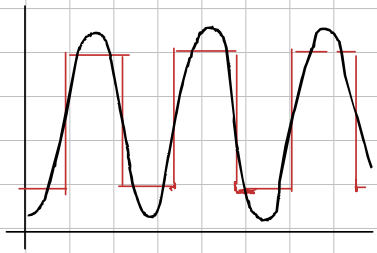
Primjer pravokutnog signala (gore)

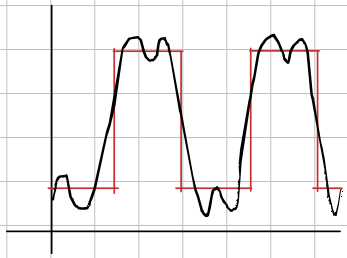
$\rightarrow$ prije su izračunati koef Fourierovog reda $F_0 = \frac{\tau}{T_0}$, $F_k = \frac{\tau}{T_0} \cdot \frac{\sin \frac{k\omega_0 \tau}{2}}{\frac{k\omega_0 \tau}{2}}$

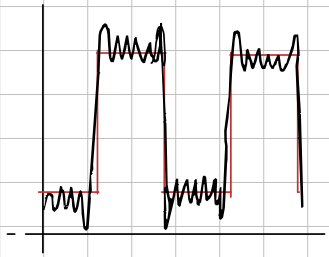
za $k = \pm 1, \pm 2 \ldots$

$\rightarrow$ aprox: $\hat{f}_k = \frac{\tau}{T_0} + \sum_{\substack{k=-K \\ k \neq 0}}^{K} \frac{\tau}{T_0} \frac{\sin \frac{k\omega_0 \tau}{2}}{\frac{k\omega_0 \tau}{2}} e^{jk\omega_0 t}$

grafovi aprox signala s konačnim brojem harmonika







povećanjem K dobivamo bolje preklapanje na horizontali, ali veća nadvišenja